Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech | ***Time: 03 Hours***

Code:101401 | ***Subject: Civil Engineering-Societal & Global Impact*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

(iv) Question No. 1 is compulsory.

Q.1 **Choose the correct option / answer the following (Any seven question only):** **[2 x 7 = 14]**

(a) Which of the following is a traditional method of solid waste management?

(i) Composting (iii) Recycling

(ii) Waste-to-energy (iv) Landfilling

(b) Which of the following methods is used for hazardous waste treatment?

(i) Incineration (iii) Landfilling

(ii) All of the above (iv) Chemical neutralization

(c) What is the main function of dams in flood control?

(i) Generating electricity (iii) Providing water for irrigation

(ii) Storing water to control downstream flooding (iv) Diverting water to reservoirs

(d) How does GIS help in disaster management?

(i) By predicting weather events (iii) By mapping disaster-prone areas

(ii) By monitoring air pollution (

(e) Which of the following was a negative Revolution?

(i) Increased soil fertility (iii) Excessive use of chemical fertilizers and pesticides

(ii) Decrease in deforestation (iv) Improved water conservation

(f) By how much has atmospheric carbon dioxide concentration increased ever since the Industrial Revolution?

(i) 20% (iii) 40%

(ii) 10% (iv) 60%

(g) What is the main purpose of a railway signaling system?

(i) To prevent train accidents (iii) To provide communication between stations

(ii) To monitor train speed (iv) To generate electricity

(h) The structural concrete slab panels can be repaired by _____

(i) Internal bonding (iii) Stress bonding

(ii) External bonding (iv) Layered bonding

(i) The preparation of the subgrade does not involve

(i) Grading (iii) Compaction

(ii) Site clearance (iv) Spreading

(j) Bituminous material is used in highway construction primarily because of them

(i) Cementing and waterproof properties (iii) Load-bearing capacity

(ii) High specific gravity (iv) Black colour which facilitates

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Q.2 (a) What are multi-purpose water projects, and why are they important? [7]
(b) What are the methods for the conservation, repair, and rehabilitation of structures and heritage buildings? [7]

Q.3 (a) What are intelligent/smart buildings, and how do they enhance sustainability? [7]
(b) What is sound control in built environments, and why is it important? [7]

Q.4 (a) Define LEED ratings and EIA. What is hyper loops? How it can play a role in futuristic growth of nation. [7]
(b) Define Ocean thermal energy conversion. Why it is in news now a days? [7]

Q.5 (a) What are the benefits of using Lean Construction in civil engineering projects? [7]
(b) How can civil engineering projects reduce their carbon footprint? [7]

Q.6 (a) Recently which energy is declared as renewable energy by GOI? How it can have impact on the total energy generation of country? [7]
(b) How does the infrastructure development & growth of the nation can have effects on employments and living standards of people? [7]

Q.7 (a) What are the major environmental concerns in modern times? [7]
(b) What role do policies play in environmental management? [7]

Q.8 (a) Examine the role of GIS in modern urban planning and environmental monitoring. [7]
(b) Enlist the modern engineering marvels of India. Describe any one of them in detail. [7]

Q.9 Write Short notes on ***any two*** of the following: [7 × 2 = 14]
(a) Various goals of sustainable development
(b) Environment Protection Acts
(c) Global warming
(d) Green Infrastructure

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